

Лабораторная работа №02

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

Загрузим нашу операционную систему и перейдем в рабочий каталог с проектом (рис. @fig-1):

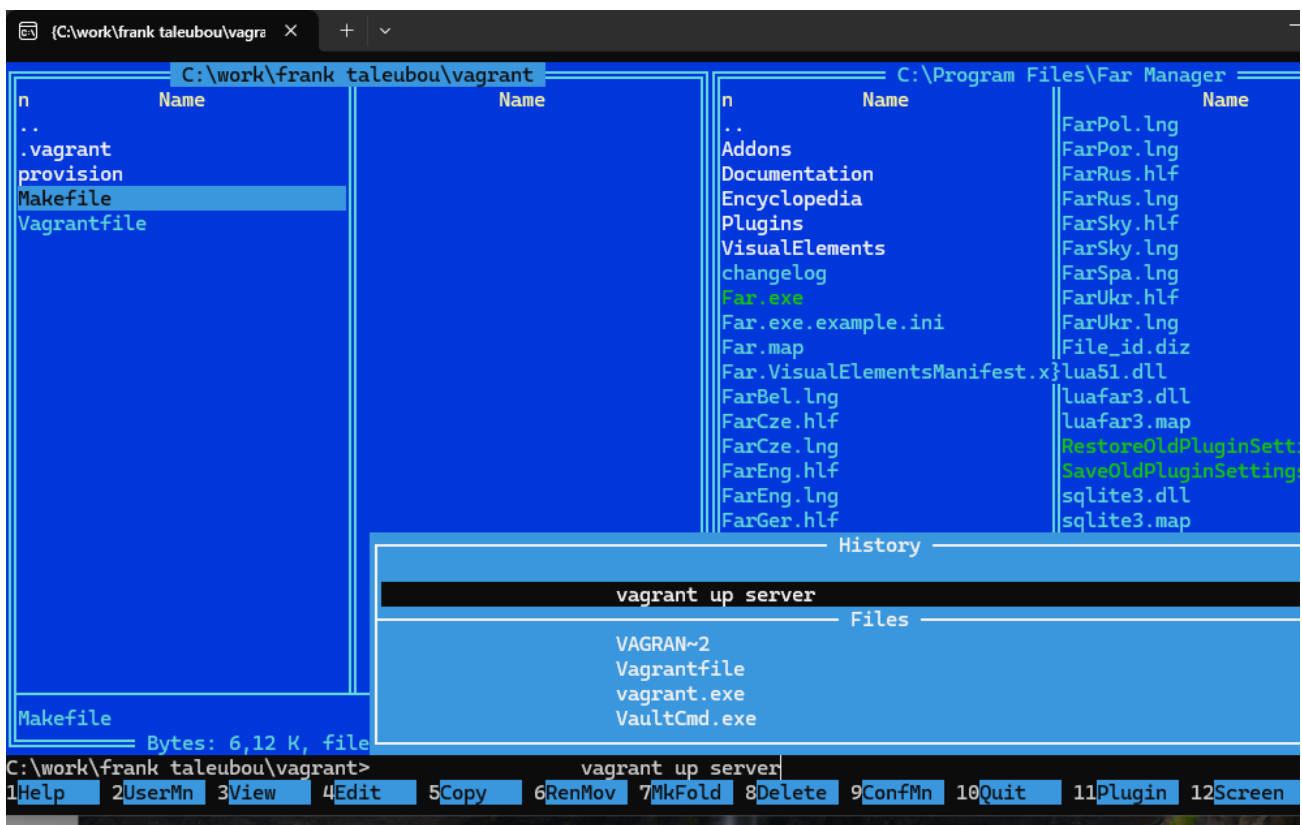


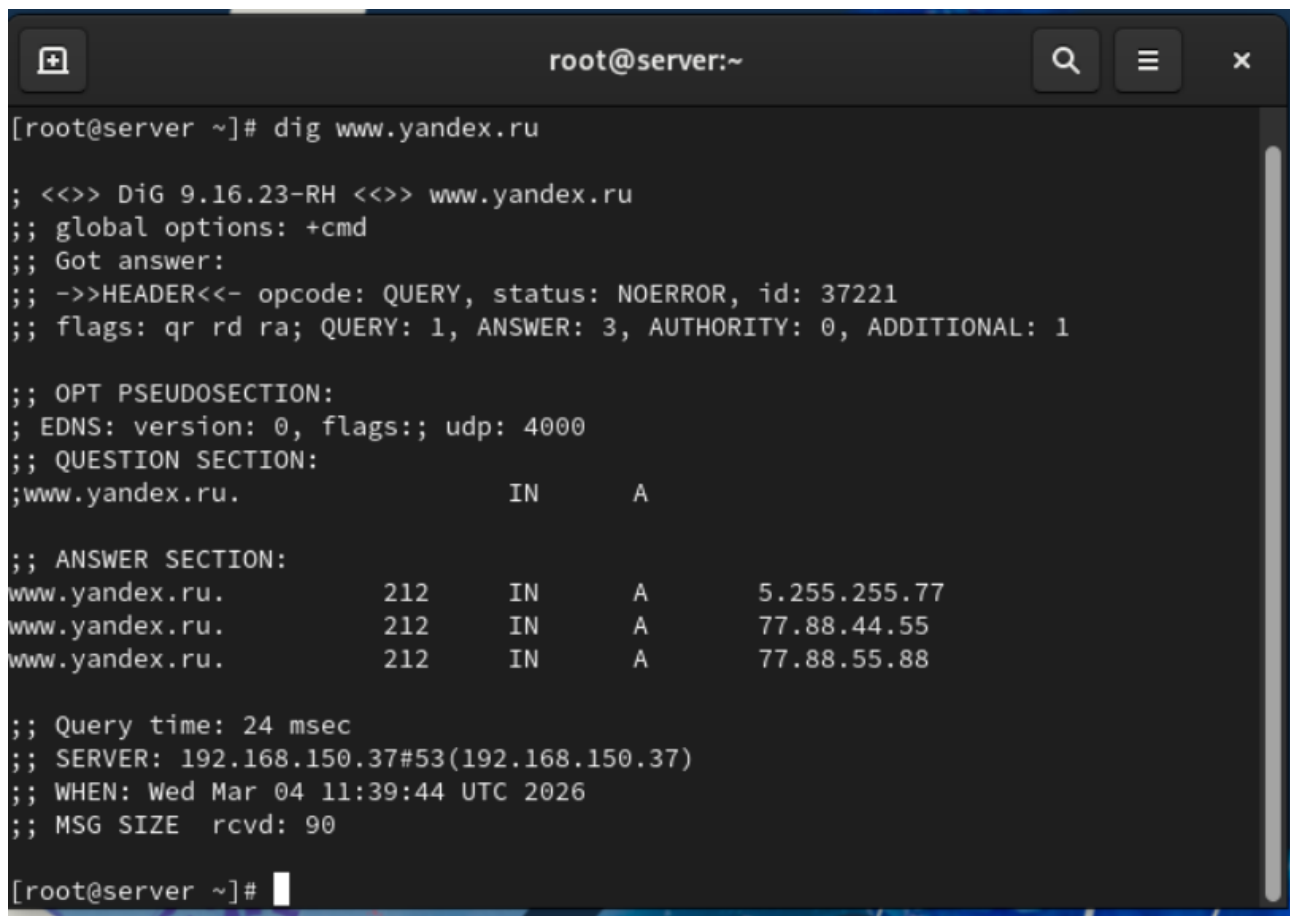
Рис. 1: Этап 1

Далее запустим виртуальную машину server (рис. @fig-2):

```
root@server:~  
Verifying      : bind-license-32:9.16.23-34.el9_7.1.noarch  
Verifying      : bind-license-32:9.16.23-18.el9_4.1.noarch  
Verifying      : bind-utils-32:9.16.23-34.el9_7.1.x86_64  
Verifying      : bind-utils-32:9.16.23-18.el9_4.1.x86_64  
Verifying      : openssl-devel-1:3.5.1-7.el9_7.x86_64  
Verifying      : openssl-devel-1:3.0.7-27.el9.x86_64  
  
Upgraded:  
  bind-libs-32:9.16.23-34.el9_7.1.x86_64  
  bind-license-32:9.16.23-34.el9_7.1.noarch  
  bind-utils-32:9.16.23-34.el9_7.1.x86_64  
  openssl-1:3.5.1-7.el9_7.x86_64  
  openssl-devel-1:3.5.1-7.el9_7.x86_64  
  openssl-libs-1:3.5.1-7.el9_7.x86_64  
Installed:  
  bind-32:9.16.23-34.el9_7.1.x86_64  
  bind-dnssec-doc-32:9.16.23-34.el9_7.1.noarch  
  bind-dnssec-utils-32:9.16.23-34.el9_7.1.x86_64  
  openssl-fips-provider-1:3.5.1-7.el9_7.x86_64  
  python3-bind-32:9.16.23-34.el9_7.1.noarch  
  python3-ply-3.11-14.el9.0.1.noarch  
  
Complete!  
[root@server ~]#
```

Рис. 2: Этап 2

На виртуальной машине server войдём под созданным нами в предыдущей работе пользователем и откроем терминал. Перейдём в режим суперпользователя и установим bind и bind-utils (рис. @fig-3):



```
root@server:~  
[root@server ~]# dig www.yandex.ru  
  
; <<>> DiG 9.16.23-RH <<>> www.yandex.ru  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37221  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 4000  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; ANSWER SECTION:  
www.yandex.ru.                212     IN      A      5.255.255.77  
www.yandex.ru.                212     IN      A      77.88.44.55  
www.yandex.ru.                212     IN      A      77.88.55.88  
  
;; Query time: 24 msec  
;; SERVER: 192.168.150.37#53(192.168.150.37)  
;; WHEN: Wed Mar 04 11:39:44 UTC 2026  
;; MSG SIZE rcvd: 90  
  
[root@server ~]#
```

Рис. 3: Этап 3

С помощью утилиты dig сделаем запрос к DNS-адресу www.yandex.ru (рис. @fig-4):

Посмотрим содержание основных конфигурационных файлов DNS (рис. @fig-5, @fig-6, @fig-7, @fig-8, @fig-9):

Просмотр конфигурационных файлов

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```
root@server:~  
[root@server ~]# dig www.yandex.ru  
  
; <<>> DiG 9.16.23-RH <<>> www.yandex.ru  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37221  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
;; EDNS: version: 0, flags:; udp: 4000  
;; QUESTION SECTION:  
;www.yandex.ru.                IN      A  
  
;; ANSWER SECTION:  
www.yandex.ru.                212     IN      A      5.255.255.77  
www.yandex.ru.                212     IN      A      77.88.44.55  
www.yandex.ru.                212     IN      A      77.88.55.88  
  
;; Query time: 24 msec  
;; SERVER: 192.168.150.37#53(192.168.150.37)  
;; WHEN: Wed Mar 04 11:39:44 UTC 2026  
;; MSG SIZE rcvd: 90  
  
[root@server ~]#
```

Рис. 4: Этап 4

```
[root@server ~]# nmcli connection edit System\ eth0  
  
===| nmcli interactive connection editor |===  
  
Editing existing '802-3-ethernet' connection: 'System eth0'  
  
Type 'help' or '?' for available commands.  
Type 'print' to show all the connection properties.  
Type 'describe [<setting>.<prop>]' for detailed property description.  
  
You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-  
1x, dcb, sriov, ethtool, match, ipv4, ipv6, hostname, link, tc, proxy  
nmcli> remove ipv4.dns  
nmcli> set ipv4.ignore-auto-dns yes  
nmcli> set ipv4.dns 127.0.0.1  
nmcli> save  
Connection 'System eth0' (5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03) successfully upd  
ated.  
nmcli> quit  
[root@server ~]#
```

Рис. 5: Этап 5

```
[root@server ~]# cat /etc/resolv.conf  
# Generated by NetworkManager  
search user.net  
nameserver 127.0.0.1  
[root@server ~]#
```

Рис. 6: Этап 6

```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory      "/var/named";
14     dump-file       "/var/named/data/cache_dump.db";
15     statistics-file "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file   "/var/named/data/named.secroots";
18     recursing-file  "/var/named/data/named.recursing";
19     allow-query     { localhost; };
20
21     /*
22     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23     - If you are building a RECURSIVE (caching) DNS server, you need to enable
24     recursion.
25     - If your recursive DNS server has a public IP address, you MUST enable access
```

Рис. 7: Этап 7

```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory      "/var/named";
14     dump-file       "/var/named/data/cache_dump.db";
15     statistics-file "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file   "/var/named/data/named.secroots";
18     recursing-file  "/var/named/data/named.recursing";
19     allow-query     { localhost; 192-168.8.8/16; };
20
21     /*
22     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23     - If you are building a RECURSIVE (caching) DNS server, you need to enable
24     recursion.
25     - If your recursive DNS server has a public IP address, you MUST enable access
```

Рис. 8: Этап 8

```
[root@server ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 127.0.0.1
[root@server ~]# firewall-cmd --add-service=dns
success
[root@server ~]# firewall-cmd --add-service=dns --permanent
success
[root@server ~]# sof | grep UDP
bash: sof: command not found...
[root@server ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
Output information may be incomplete.
```

Рис. 9: Этап 9

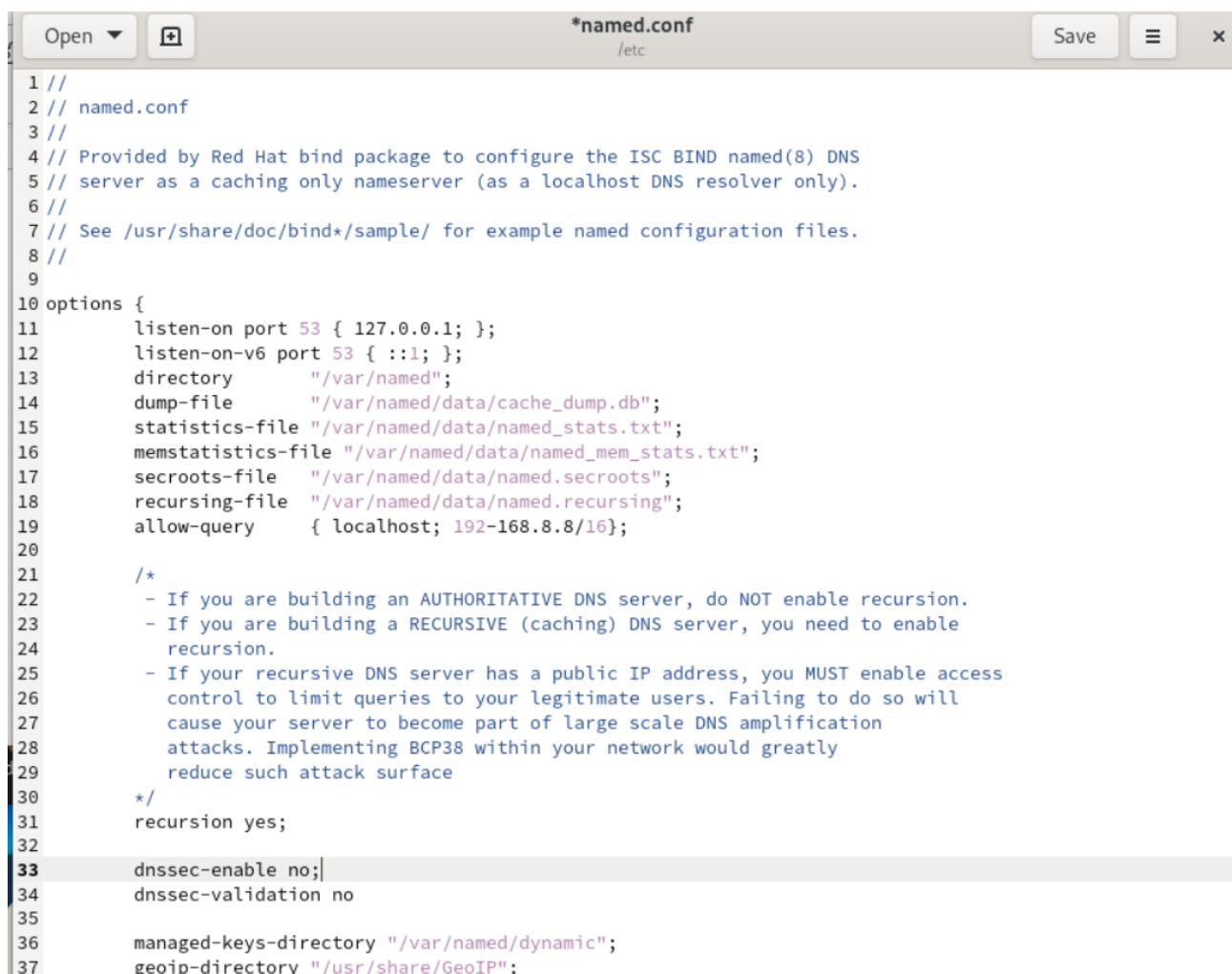
Запустим DNS-сервер и включим его в автозагрузку (рис. @fig-10):

```
named 3187 3240 isc-net-0 named 67u IPv6 34440 0t0 UDP lo
calhost:domain
named 3187 3240 isc-net-0 named 68u IPv6 34441 0t0 UDP lo
calhost:domain
named 3187 3240 isc-net-0 named 69u IPv6 34442 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 54u IPv4 33481 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 55u IPv4 33482 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 56u IPv4 33483 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 57u IPv4 33484 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 66u IPv6 34439 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 67u IPv6 34440 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 68u IPv6 34441 0t0 UDP lo
calhost:domain
named 3187 3241 isc-net-0 named 69u IPv6 34442 0t0 UDP lo
calhost:domain
NetworkMa 3312 root 27u IPv4 33731 0t0 UDP se
rver.user.net:bootpc->_gateway:bootps
NetworkMa 3312 3320 gmain root 27u IPv4 33731 0t0 UDP se
rver.user.net:bootpc->_gateway:bootps
NetworkMa 3312 3322 gdbus root 27u IPv4 33731 0t0 UDP se
rver.user.net:bootpc->_gateway:bootps
[root@server ~]#
```

Рис. 10: Этап 10

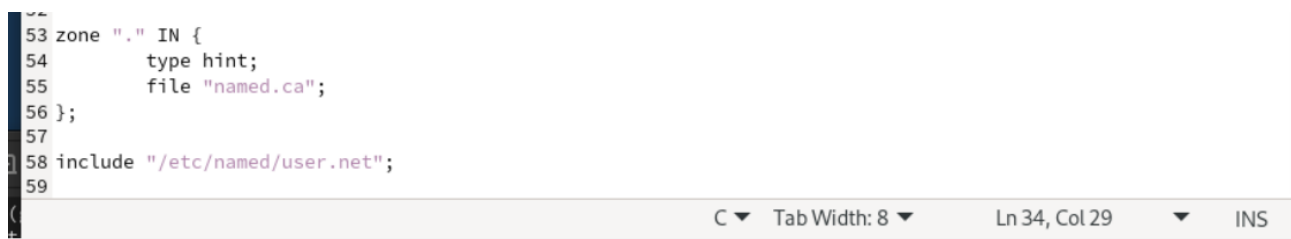
Проанализируем отличие в выведенной на экран информации при выполнении команд (рис. @fig-11):

Сделаем DNS-сервер сервером по умолчанию для хоста server и внутренней виртуальной сети. Для этого изменим настройки сетевого соединения eth0 в NetworkManager (рис. @fig-12):



```
1 //
2 // named.conf
3 //
4 // Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
5 // server as a caching only nameserver (as a localhost DNS resolver only).
6 //
7 // See /usr/share/doc/bind*/sample/ for example named configuration files.
8 //
9
10 options {
11     listen-on port 53 { 127.0.0.1; };
12     listen-on-v6 port 53 { ::1; };
13     directory "/var/named";
14     dump-file "/var/named/data/cache_dump.db";
15     statistics-file "/var/named/data/named_stats.txt";
16     memstatistics-file "/var/named/data/named_mem_stats.txt";
17     secroots-file "/var/named/data/named.secroots";
18     recursing-file "/var/named/data/named.recursing";
19     allow-query { localhost; 192-168.8.8/16};
20
21     /*
22      - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
23      - If you are building a RECURSIVE (caching) DNS server, you need to enable
24        recursion.
25      - If your recursive DNS server has a public IP address, you MUST enable access
26        control to limit queries to your legitimate users. Failing to do so will
27        cause your server to become part of large scale DNS amplification
28        attacks. Implementing BCP38 within your network would greatly
29        reduce such attack surface
30     */
31     recursion yes;
32
33     dnssec-enable no;
34     dnssec-validation no
35
36     managed-keys-directory "/var/named/dynamic";
37     geoip-directory "/usr/share/GeoIP";
```

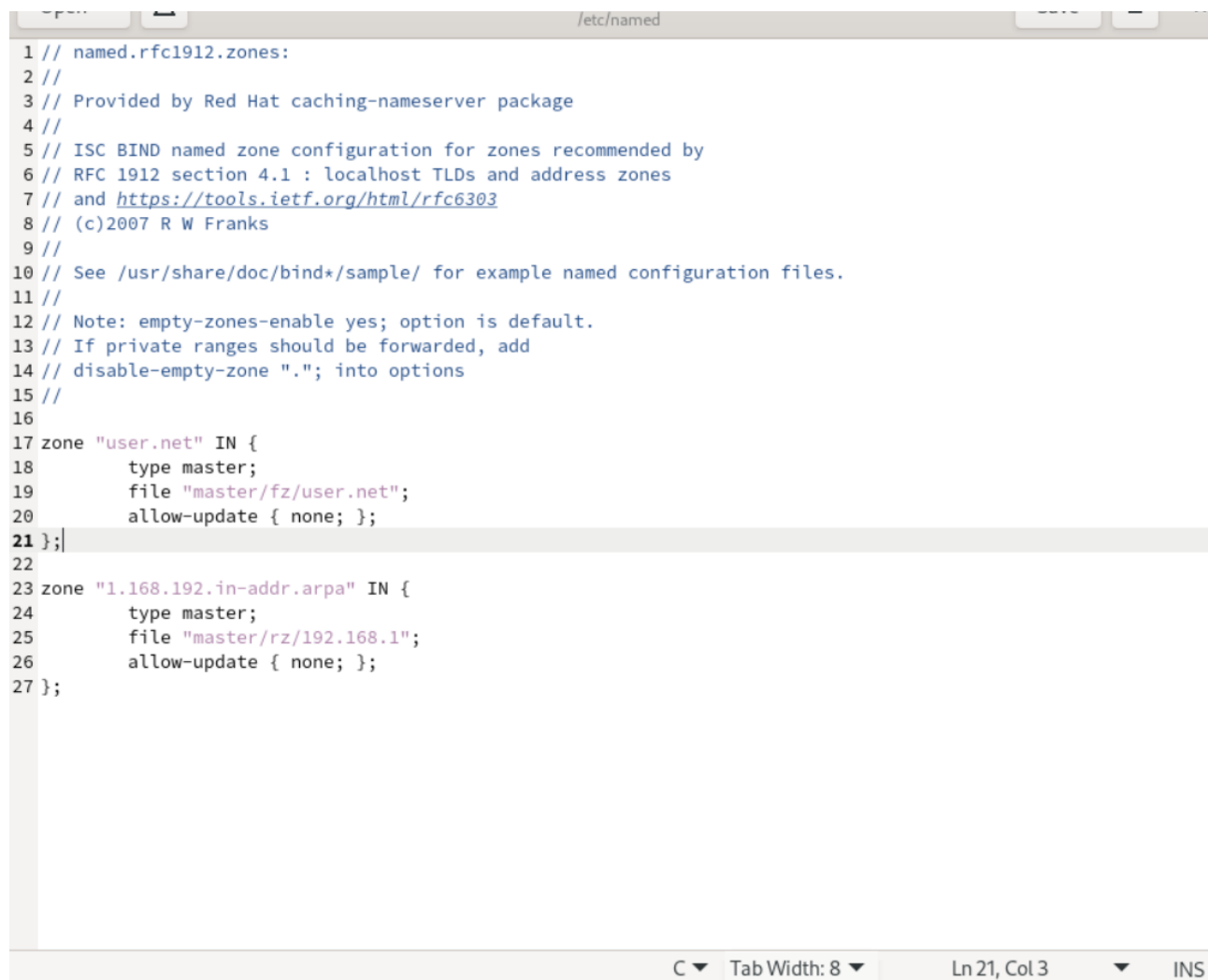
Рис. 11: Этап 11



```
53 zone "." IN {
54     type hint;
55     file "named.ca";
56 };
57
58 include "/etc/named/user.net";
59
```

Рис. 12: Этап 12

Повторим действия для соединения System eth0 (рис. @fig-13):



```
1 // named.rfc1912.zones:
2 //
3 // Provided by Red Hat caching-nameserver package
4 //
5 // ISC BIND named zone configuration for zones recommended by
6 // RFC 1912 section 4.1 : localhost TLDs and address zones
7 // and https://tools.ietf.org/html/rfc6303
8 // (c)2007 R W Franks
9 //
10 // See /usr/share/doc/bind*/sample/ for example named configuration files.
11 //
12 // Note: empty-zones-enable yes; option is default.
13 // If private ranges should be forwarded, add
14 // disable-empty-zone "."; into options
15 //
16
17 zone "user.net" IN {
18     type master;
19     file "master/fz/user.net";
20     allow-update { none; };
21 };
22
23 zone "1.168.192.in-addr.arpa" IN {
24     type master;
25     file "master/rz/192.168.1";
26     allow-update { none; };
27 };
```

Рис. 13: Этап 13

Перезапустим NetworkManager и проверим наличие изменений в файле /etc/resolv.conf (рис. @fig-14):

Внесём изменения в файл /etc/named.conf для настройки направления DNS-запросов (рис. @fig-15):

Внесём изменения в настройки межсетевого экрана узла server, разрешив работу с DNS (рис. @fig-16):

Добавим перенаправление DNS-запросов на вышестоящий DNS-сервер и отключим DNSSEC при необходимости (рис. @fig-17):

Откроем файл /etc/named/user.net и пропишем прямую и обратную зоны (рис. @fig-18):


```
root@server:/var/named/master/rz

(gedit:4392): dconf-WARNING **: 12:32:52.732: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4392): dconf-WARNING **: 12:32:52.940: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4392): dconf-WARNING **: 12:32:52.946: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

* (gedit:4392): WARNING **: 12:38:33.803: Set document metadata failed: Setting
attribute metadata::gedit-spell-language not supported

* (gedit:4392): WARNING **: 12:38:33.809: Set document metadata failed: Setting
attribute metadata::gedit-encoding not supported

* (gedit:4392): WARNING **: 12:40:35.774: Set document metadata failed: Setting
attribute metadata::gedit-position not supported

(gedit:4392): dconf-WARNING **: 12:40:35.795: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)
root@server etc]# cp /var/named/named.loopback /var/named/master/rz/
root@server etc]# cd /var/named/master/rz/
root@server rz]# mv named.loopback 192.168.1
root@server rz]#
```

Рис. 16: Этап 16

```
*192.168.1
/var/named/master/rz/var/named/master/rz

1 $TTL 1D
2 @ IN SOA      @server.user.net. (
3             2024072700 ; seria
4             1D ; refresh
5             1H ; retry
6             1W ; expire
7             3H ) ; minimum
8             NS @
9             A 192.168.1.1
10            PTR server.user.net.
11            $ORIGIN 1.168.192.in-addr.arpa.
12            1 PTR server.user.net.
13            1 PTR ns.user.net.
```

Рис. 17: Этап 17

```

(gedit:4392): dconf-WARNING **: 12:40:35.795: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)
[root@server etc]# cp /var/named/named.loopback /var/named/master/rz/
[root@server etc]# cd /var/named/master/rz/
[root@server rz]# mv named.loopback 192.168.1
[root@server rz]# chown -R named:named /etc/named
[root@server rz]# chown -R named:named /var/named
[root@server rz]# restorecon -vR /etc
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:u
ser_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/named.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:ob
ject_r:named_conf_t:s0
[root@server rz]# restorecon -vR /var/named
[root@server rz]#

```

Рис. 18: Этап 18

Создадим подкаталоги для файлов зон (рис. @fig-19):

```

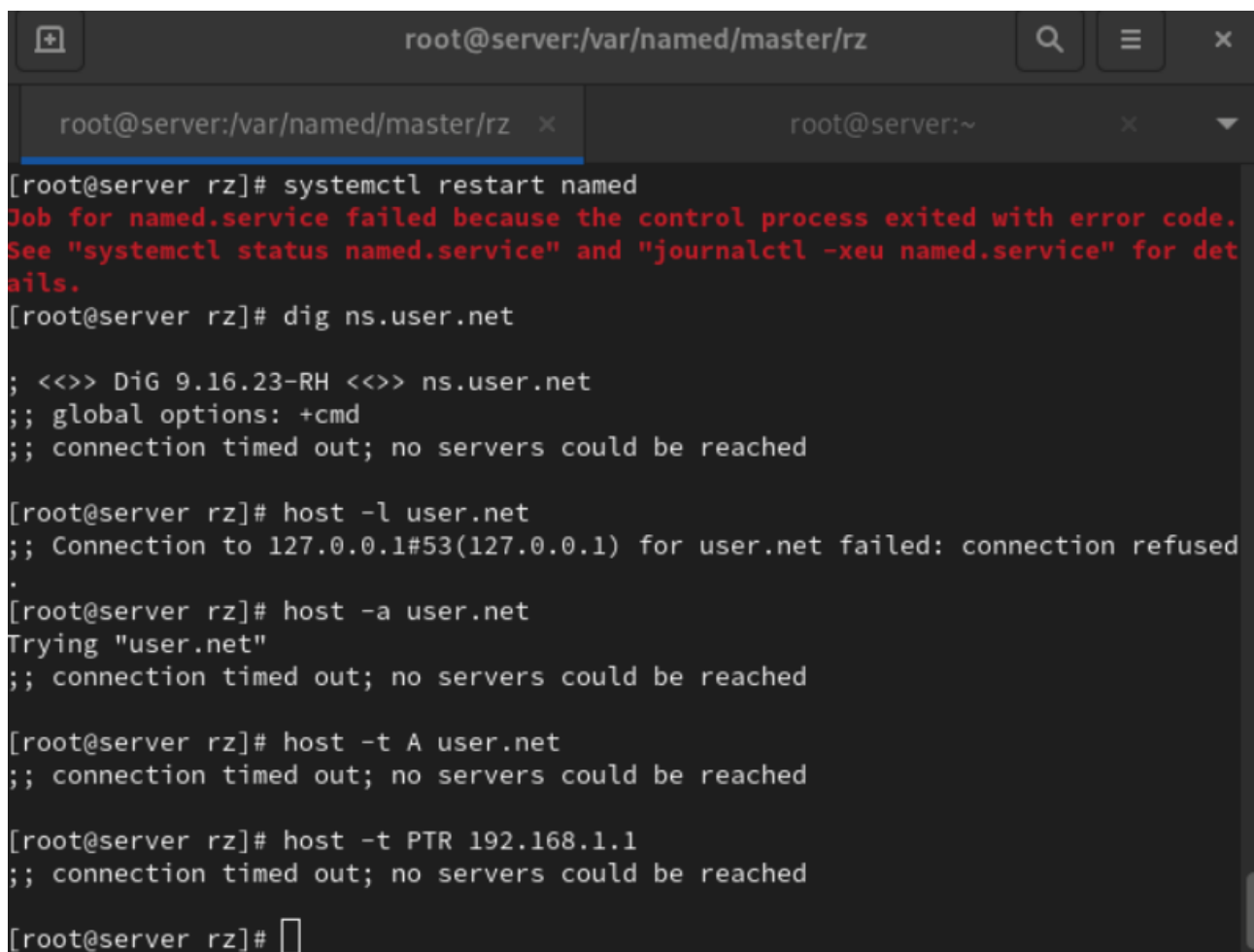
root@server:~
root@server:/var/named/master/rz x root@server:~ x
bash: journalctl: command not found...
[root@server ~]# journalctl -x -f
Mar 04 12:55:21 server.user.net systemd[1586]: Started VTE child process 4576 la
unched by gnome-terminal-server process 2556.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support
A start job for unit UNIT has finished successfully.
The job identifier is 679.
Mar 04 12:56:07 server.user.net sudo[4613]:  vagrant : TTY=pts/2 ; PWD=/root ; U
SER=root ; COMMAND=/bin/bash
Mar 04 12:56:07 server.user.net dbus-broker-launch[637]: avc:  op=load_policy ls
m=selinux seqno=4 res=1
Mar 04 12:56:07 server.user.net sudo[4613]: pam_unix(sudo-i:session): session op
ened for user root(uid=0) by vagrant(uid=1000)
Mar 04 12:56:07 server.user.net systemd[1]: Starting Hostname Service...
Subject: A start job for unit systemd-hostnamed.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support
A start job for unit systemd-hostnamed.service has begun execution.

```

Рис. 19: Этап 19

Скопируем шаблон прямой DNS-зоны и переименуем его (рис. @fig-20):

Отредактируем файл прямой зоны (рис. @fig-21):



A terminal window titled 'root@server:/var/named/master/rz' with a search icon, menu icon, and close icon in the title bar. The terminal shows the following commands and output:

```
[root@server rz]# systemctl restart named
Job for named.service failed because the control process exited with error code.
See "systemctl status named.service" and "journalctl -xeu named.service" for details.
[root@server rz]# dig ns.user.net

; <<>> DiG 9.16.23-RH <<>> ns.user.net
;; global options: +cmd
;; connection timed out; no servers could be reached

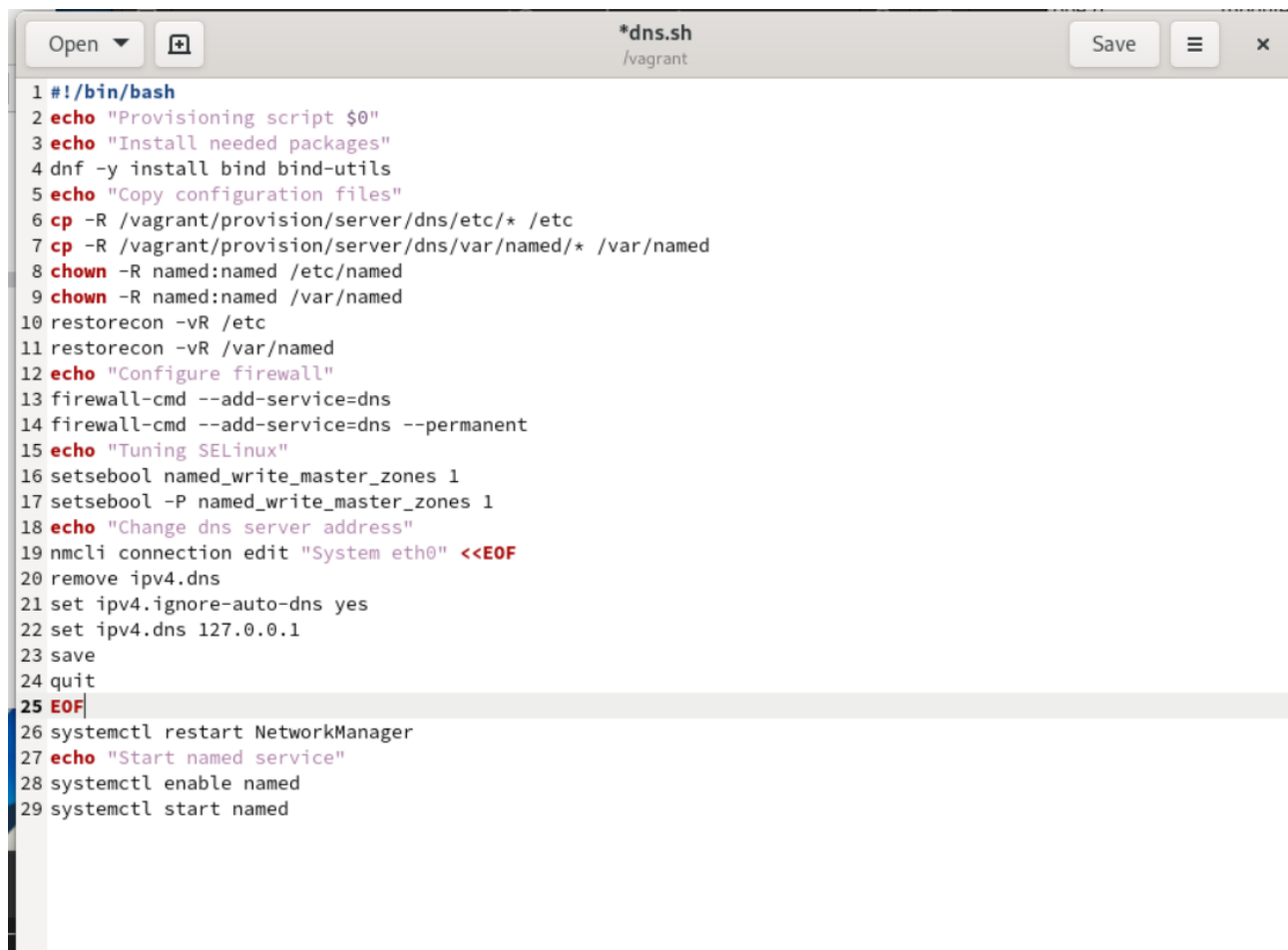
[root@server rz]# host -l user.net
;; Connection to 127.0.0.1#53(127.0.0.1) for user.net failed: connection refused
.
[root@server rz]# host -a user.net
Trying "user.net"
;; connection timed out; no servers could be reached

[root@server rz]# host -t A user.net
;; connection timed out; no servers could be reached

[root@server rz]# host -t PTR 192.168.1.1
;; connection timed out; no servers could be reached

[root@server rz]#
```

Рис. 20: Этап 20

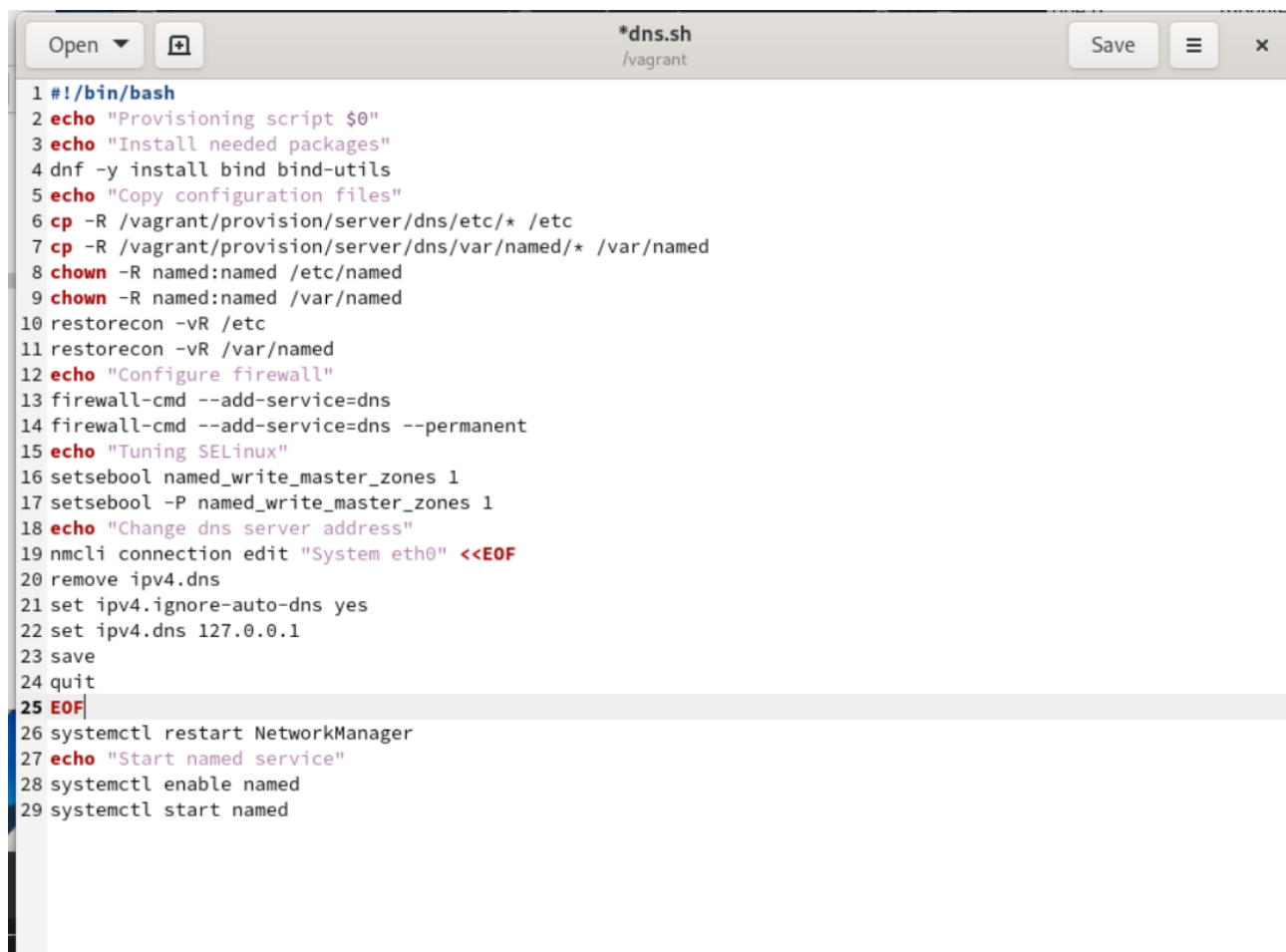


A script editor window titled '*dns.sh /vagrant' with 'Open', 'Save', menu icon, and close icon in the title bar. The script content is as follows:

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install bind bind-utils
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/dns/etc/* /etc
7 cp -R /vagrant/provision/server/dns/var/named/* /var/named
8 chown -R named:named /etc/named
9 chown -R named:named /var/named
10 restorecon -vR /etc
11 restorecon -vR /var/named
12 echo "Configure firewall"
13 firewall-cmd --add-service=dns
14 firewall-cmd --add-service=dns --permanent
15 echo "Tuning SELinux"
16 setsebool named_write_master_zones 1
17 setsebool -P named_write_master_zones 1
18 echo "Change dns server address"
19 nmcli connection edit "System eth0" <<EOF
20 remove ipv4.dns
21 set ipv4.ignore-auto-dns yes
22 set ipv4.dns 127.0.0.1
23 save
24 quit
25 EOF
26 systemctl restart NetworkManager
27 echo "Start named service"
28 systemctl enable named
29 systemctl start named
```

Рис. 21: Этап 21

Скопируем шаблон обратной DNS-зоны и переименуем его (рис. @fig-22):



```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install bind bind-utils
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/dns/etc/* /etc
7 cp -R /vagrant/provision/server/dns/var/named/* /var/named
8 chown -R named:named /etc/named
9 chown -R named:named /var/named
10 restorecon -vR /etc
11 restorecon -vR /var/named
12 echo "Configure firewall"
13 firewall-cmd --add-service=dns
14 firewall-cmd --add-service=dns --permanent
15 echo "Tuning SELinux"
16 setsebool named_write_master_zones 1
17 setsebool -P named_write_master_zones 1
18 echo "Change dns server address"
19 nmcli connection edit "System eth0" <<EOF
20 remove ipv4.dns
21 set ipv4.ignore-auto-dns yes
22 set ipv4.dns 127.0.0.1
23 save
24 quit
25 EOF
26 systemctl restart NetworkManager
27 echo "Start named service"
28 systemctl enable named
29 systemctl start named
```

Рис. 22: Этап 22

Отредактируем файл обратной зоны (рис. @fig-23):

Исправим права доступа к файлам и восстановим метки SELinux (рис. @fig-24):

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```
[root@server ~]# dig @127.0.0.1 www.yandex.ru

; <<>> DiG 9.16.23-RH <<>> @127.0.0.1 www.yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 25062
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1232
; COOKIE: ef4e057ac4fd09100100000069a81aa90d91fa158f9ab9ef (good)
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; Query time: 826 msec
;; SERVER: 127.0.0.1#53(127.0.0.1)
;; WHEN: Wed Mar 04 11:42:33 UTC 2026
;; MSG SIZE rcvd: 70

[root@server ~]#
```

Рис. 23: Этап 23

```
[root@server ~]# nmcli connection edit eth0

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-
1x, dcb, sriov, ethtool, match, ipv4, ipv6, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (52b757e4-3833-4753-b0eb-de0af7f26f57) successfully updated.
nmcli> quit
[root@server ~]#
```

Рис. 24: Этап 24